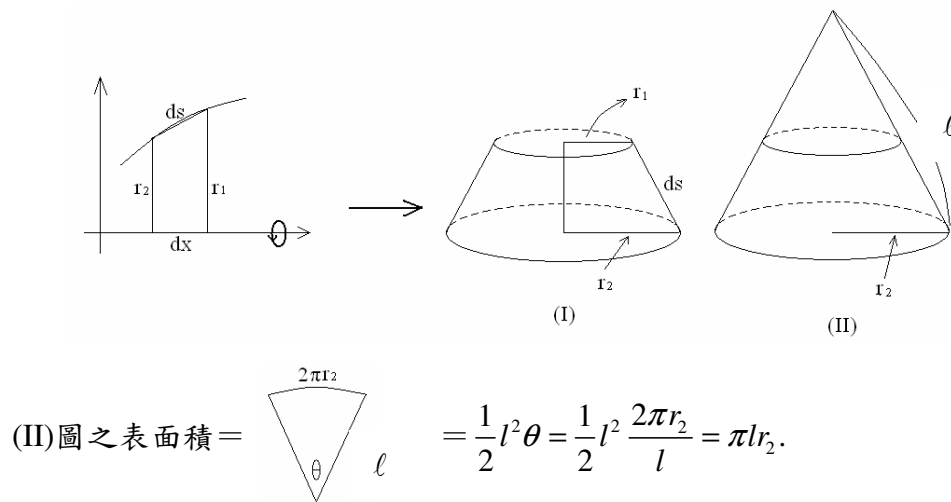


§8-2 Area of a Surface Revolution

Homework : 5,11,29,33,35

公式：Area of a Surface Revolution = S



$$(I) \text{圖之表面積} = \pi(l_1 + l_2)r_2 - \pi l_1 r_1 = \pi[(r_2 - r_1)l_1 + r_2 l_2]$$

$$\begin{aligned} \frac{l_1}{l_1 + l_2} &= \frac{r_1}{r_2} \\ (r_2 - r_1)l_1 &= r_1 l_2 \end{aligned}$$

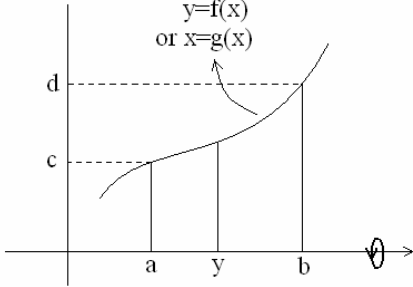
$$\therefore (I) \text{圖之表面積} = \pi(r_1 + r_2)l_2$$

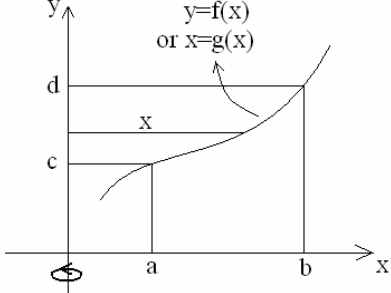
$$= \pi(r_1 + r_2)ds$$

$$= 2\pi r ds$$

$$r = \frac{r_1 + r_2}{2} (= y = f(x) \text{ 當 } dx \text{ 是無窮小時}).$$

$$\Rightarrow S = \int 2\pi r ds.$$

$$S = \int 2\pi r ds = \begin{cases} \int_a^b 2\pi f(x) \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx \\ \int_c^d 2\pi y \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy. \end{cases}$$


$$S = \int 2\pi r ds = \begin{cases} \int_a^b 2\pi x \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx \\ \int_c^d 2\pi g(y) \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy. \end{cases}$$


Example 1 : $y = \sqrt{r^2 - x^2}$, $-r \leq x \leq r$; the ax -axis .

(球的表面積)

Solution :

$$y' = (-x)(r^2 - x^2)^{-\frac{1}{2}} \Rightarrow 1 + (y')^2 = \frac{r^2}{r^2 - x^2}$$

$$\begin{aligned} S &= \int_{-r}^r 2\pi \sqrt{r^2 - x^2} \frac{r}{\sqrt{r^2 - x^2}} dx \\ &= 2\pi r \int_{-r}^r dx \\ &= 4\pi r^2. \end{aligned}$$

Example 2 : $y = x^2$, (1,1) to (2,4) ; the y -axis .

Solution :

$$\begin{aligned} x &= y^{\frac{1}{2}} \\ \Rightarrow x' &= \frac{1}{2} y^{-\frac{1}{2}} \\ \Rightarrow 1 + (x')^2 &= \frac{1 + 4y}{4y} \end{aligned}$$

$$\begin{aligned}
S &= \int_1^4 2\pi y^{\frac{1}{2}} \frac{\sqrt{1+4y}}{2\sqrt{y}} dy \\
&= \pi \int_1^4 (1+4y)^{\frac{1}{2}} dy \\
&= 2\pi \frac{2}{3} \left(y + \frac{1}{4} \right)^{\frac{3}{2}} \Big|_1^4 \\
&= \frac{\pi}{6} (17\sqrt{17} - 5\sqrt{5}).
\end{aligned}$$

註：另一解法為：

$$\begin{aligned}
y' &= 2x \Rightarrow 1 + (y')^2 = 1 + 4x^2 \\
S &= \int_1^2 2\pi x \sqrt{1+4x^2} dx \quad 1 \\
&= \frac{\pi}{6} (17\sqrt{17} - 5\sqrt{5}).
\end{aligned}$$

Example 3 : $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$; the x -axis.

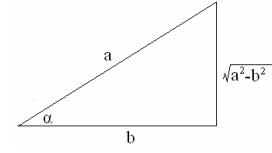
(橢圓體的表面積)

Solution :

$$\begin{aligned}
y &= \frac{b}{a} \sqrt{a^2 - x^2} \\
\Rightarrow y' &= \frac{b}{a} (-x) (a^2 - x^2)^{-\frac{1}{2}} \\
\Rightarrow 1 + (y')^2 &= \frac{b^2 x^2 + a^4 - a^2 x^2}{a^2 (a^2 - x^2)}.
\end{aligned}$$

$$\begin{aligned}
S &= 2 \int_0^a 2\pi \frac{b}{a} \sqrt{a^2 - x^2} \frac{\sqrt{a^4 - (a^2 - b^2)x^2}}{a\sqrt{a^2 - x^2}} dx \\
&= \frac{4\pi b}{a^2} \int_0^a \sqrt{a^4 - (a^2 - b^2)x^2} dx
\end{aligned}$$

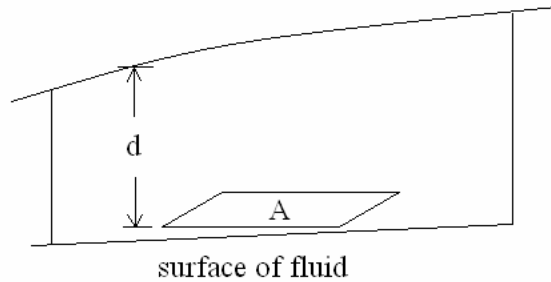
$$\begin{aligned}
x &= \frac{a^2}{\sqrt{a^2 - b^2}} \sin \theta \\
&= \frac{4\pi b}{a^2} \int_0^\alpha a^2 \cos \theta \frac{a^2}{\sqrt{a^2 - b^2}} \cos \theta d\theta \\
&= \frac{4\pi a^2 b}{\sqrt{a^2 - b^2}} \int_0^\alpha \cos^2 \theta d\theta \\
&= \frac{4\pi a^2 b}{\sqrt{a^2 - b^2}} \left(\frac{\theta}{2} + \frac{\sin \theta \cos \theta}{2} \Big|_0^\alpha \right) \\
&= \frac{4\pi a^2 b}{\sqrt{a^2 - b^2}} \left(\frac{1}{2} \sin^{-1} \frac{\sqrt{a^2 - b^2}}{a} + \frac{b\sqrt{a^2 - b^2}}{2a^2} \right) \\
&= 2\pi \left[b^2 + \frac{a^2 b}{\sqrt{a^2 - b^2}} \sin^{-1} \frac{\sqrt{a^2 - b^2}}{a} \right]
\end{aligned}$$



§8-3 Applications to Physics and Engineering

Homework : 1,11,27,34

(I) Hydrostatic Pressure and Force



$$p = \frac{F}{A} = \frac{mg}{A} = \frac{\rho A d}{A} = \rho g d.$$

F : force m : mass g : gravity acceleration.

ρ : density of a fluid in metric system.

δ : density of a fluid in ft-pound system.

Water density : $1000 \frac{\text{kg}}{\text{m}^3}$ or $62.5 \text{ pd}/\text{ft}^3$ (含重力)

$$(1) \quad P(\text{pressure}) = \begin{cases} \rho g d & (\text{metric system}) \\ \delta d & (\text{ft-pound system}). \end{cases}$$

Force against a vertical plate or wall or dam.

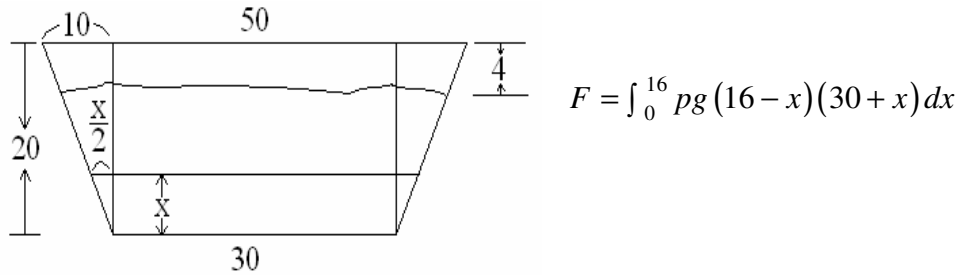
(2)



$$F = \int p dA \quad (p, A \text{ 隨位置而變})$$

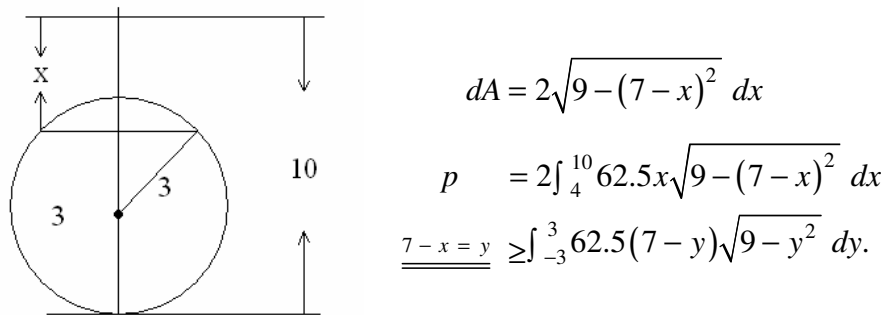
Example 1 : A dam has the shape of the trapezoid. The height is 20m, and width is 50m at the top and 30m at the bottom. Find the force on the dam due to hydrostatic pressure if the water level is 4 m from the top of the dam.

Solution :



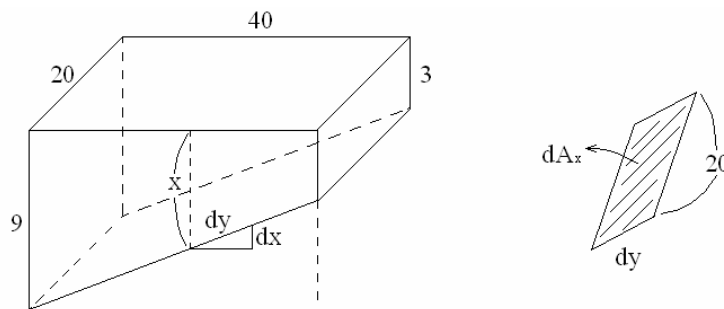
Example 2 : Find the hydro static force on one end of a cylindrical drum with radius 3 ft if the drum is submerged in water 10 ft deep.

Solution :



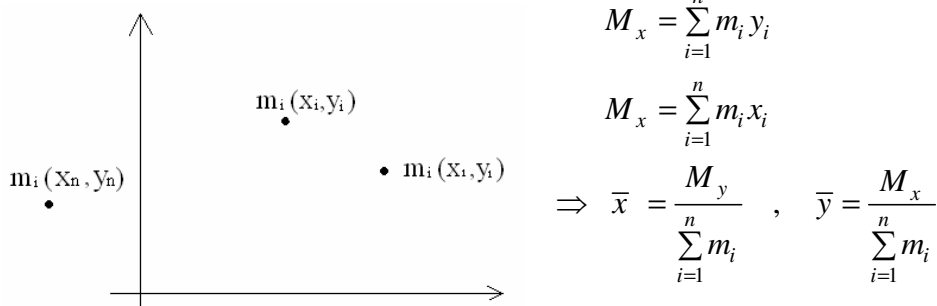
Example 3 : Force against an inclined bottom.

Solution :

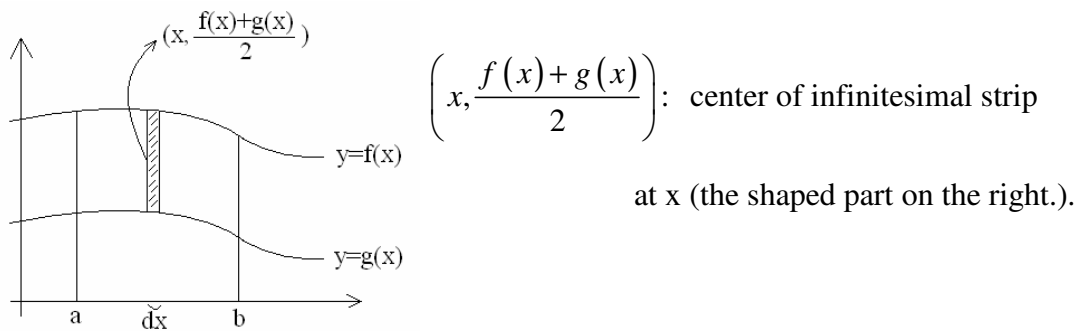


$$\begin{aligned}
& \int \frac{9}{3} \rho dA_x \\
&= \int \frac{9}{3} \delta dA_x \\
&= \int \frac{9}{3} \delta x 20 dy \\
&\frac{dy}{dx} = \frac{\sqrt{1636}}{6} = \frac{\sqrt{409}}{3} \Rightarrow dy = \frac{\sqrt{409}}{3} dx \\
&= \int \frac{9}{3} \frac{20\sqrt{409}}{3} \delta x dx.
\end{aligned}$$

Center $(\bar{x}, \bar{y})$ of a discrete system.



Center $(\bar{x}, \bar{y})$ of a continuous system with uniform density ρ .



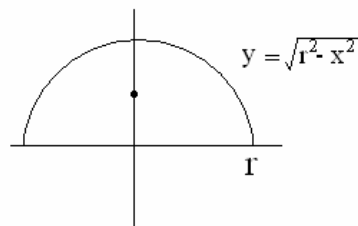
$$\begin{aligned}
M_x &= \int_a^b \frac{(f(x) + g(x))}{2} \cdot \rho(f(x) + g(x)) dx = \rho \int_a^b \frac{(f^2(x) + g^2(x))}{2} dx. \\
M_y &= \int_a^b x \cdot \rho(f(x) + g(x)) dx = \rho \int_a^b x \cdot (f(x) + g(x)) dx.
\end{aligned}$$

$$\Rightarrow \bar{y} = \frac{\frac{1}{2} \int_a^b (f^2(x) - g^2(x)) dx}{\int_a^b (f(x) - g(x)) dx}$$

$$\bar{x} = \frac{\int_a^b x(f(x) - g(x)) dx}{\int_a^b (f(x) - g(x)) dx}$$

Example 1 : Find the center of a semicircular plate of radius r .

Solution :

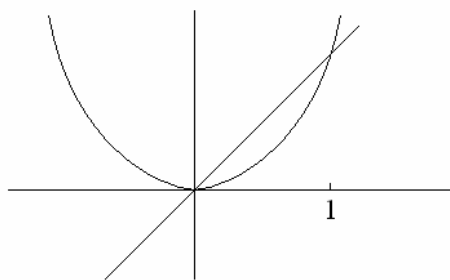


$$\Rightarrow \bar{y} = \frac{\frac{1}{2} \int_{-r}^r (\sqrt{r^2 - x^2})^2 dx}{\frac{1}{2} \pi r^2} = \frac{\int_{-r}^r (r^2 - x^2) dx}{\pi r^2}$$

$$= \frac{1}{\pi r^2} \left[r^2 x - \frac{x^3}{3} \right]_{-r}^r = \frac{4r}{3\pi}$$

Example 2 : Find the centroid of the region bounded by $y = x$ and $y = x^2$.

Solution :



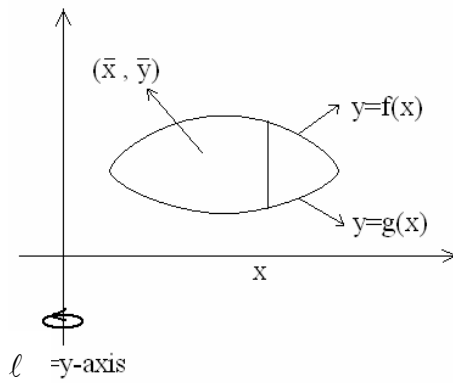
$$A = \int_0^1 (x - x^2) dx = \frac{1}{6}$$

$$M_y = \int_0^1 \frac{1}{2} (x - x^2) dx = \frac{1}{15}$$

$$M_x = \int_0^1 x(x - x^2) dx = \frac{1}{12}$$

$$\Rightarrow \bar{x} = \frac{1}{2}, \quad \bar{y} = \frac{2}{5}.$$

Theorem of Pappu :



$$\begin{aligned}
 V &= \int_a^b 2\pi x (f(x) - g(x)) dx \\
 &= 2\pi \int_a^b x (f(x) - g(x)) dx \\
 &= 2\pi \bar{x} A \\
 &= (2\pi \bar{x}) A \\
 &= (\text{distance traveled by of the region}) \times (\text{the area of the region}). \quad 2
 \end{aligned}$$